APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

MODEL QUESTION PAPER

SEMESTER - 2

B. Tech.

GROUP A

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MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
SECOND SEMESTER B. TECH DEGREE EXAMINATION, MAY 2025

Course Code: GAMAT201

Course Name: MATHEMATICS FOR COMPUTER AND INFORMATION SCIENCE – 2

Max. Marks: 60 | Duration: 2 hours 30 minutes

PART A

		Answer all questions. Each question carries 3 marks	CO	Marks
1		Write a linear system of two equations with two unknowns which have infinitely many solutions.	CO1	(3)
2		Find the rank of the matrix A= $\begin{bmatrix} 1 & -1 & 2 \\ 3 & 1 & 4 \\ 5 & -1 & 8 \end{bmatrix}$	CO1	(3)
3		Check whether W ={(x, x+3) : x ∈ R} is a subspace of R^2 under standard operations.	CO2	(3)
4		Find the coordinate matrix of u = (1, -1) relative to the non-standard basis {(1, 0), (1, 1)} in R^2.	CO2	(3)
5		Find the norm of u + v, where u = (1, -1, 3, 2) and v = (-1, 1, 0, 2).	CO3	(3)
6		Find the vector v with length 3 and in the direction of u = (1, 2, 2).	CO3	(3)
7		Let T : $R^5 \rightarrow R^7$ be a linear transformation. Find the dimension of the kernel of T when the dimension of the range is 2.	CO4	(3)
8		Let T : $R^2 \rightarrow R^2$ be a linear transformation defined by T (x, y) = (x - y, x+3y) Find the standard matrix for T.	CO4	(3)

PART B

Answer any one full question from each module. Each question carries 9 marks

Module 1

9	a)	Using Gauss elimination method solve the linear system x + y − z = 9, 8y + 6z = −6, −2x + 4y − 6z = 40.	CO1	(4)
	b)	Find the eigenvalues and eigenvectors of A = $\begin{bmatrix} 2 & 0 \\ -1 & 4 \end{bmatrix}$.	CO1	(5)
10	a)	Find the values of λ and μ for which the system of equations x + 3y -2z = 2, -x + 3y − 2z = 4, 2x + y - λz = μ has (i) no solution (ii) a unique solution (iii) a one parameter family of solutions.	CO1	(4)

<table>
<tr><td></td><td>b)</td><td>Diagonalize the matrix A=$\begin{bmatrix} 3 & 5 & 3 \\ 0 & 4 & 6 \\ 0 & 0 & 1 \end{bmatrix}$ if possible.</td><td>CO1</td><td>(5)</td></tr>
<tr><td colspan="5">Module 2</td></tr>
<tr><td>11</td><td>a)</td><td>Check whether set of all points on the plane x +y -2z = 0 is a subspace of R^3 under standard operations.</td><td>CO2</td><td>(4)</td></tr>
<tr><td></td><td>b)</td><td>Show that the set of vectors {(1, 3, −1), (2, 3, 3), (3, 0, −1)} forms a basis of R^3.</td><td>CO2</td><td>(5)</td></tr>
<tr><td>12</td><td>a)</td><td>Check whether the vector (1, −2, 2) can be written as a linear combination of the vectors in the set {(1, 2, 3), (0, 1, 2), (−1, 0, 1)}.</td><td>CO2</td><td>(4)</td></tr>
<tr><td></td><td>b)</td><td>Given B = {(1, 3), (−2, −2)} and B′ = {(−12, 0), (−4, 4)} are two bases of R^2. Find the transition matrix from B′ to B.</td><td>CO2</td><td>(5)</td></tr>
<tr><td colspan="5">Module 3</td></tr>
<tr><td>13</td><td>a)</td><td>Verify the Cauchy-Schwarz inequality for A =$\begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}$, B = $\begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix}$ with inner product $\langle A, B\rangle = a_{11}b_{11} + a_{12}b_{12} + a_{21}b_{21} + a_{22}b_{22}$.</td><td>CO3</td><td>(4)</td></tr>
<tr><td></td><td>b)</td><td>Find the least squares regression line for the following data
<table>
<tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr>
<tr><td>y</td><td>2</td><td>3</td><td>5</td><td>9</td><td>7</td></tr>
</table></td><td>CO3</td><td>(5)</td></tr>
<tr><td>14</td><td>a)</td><td>Find the orthogonal projection of u = (1, -2) onto v = (3, 2) in R^2.</td><td>CO3</td><td>(3)</td></tr>
<tr><td></td><td>b)</td><td>Find the orthonormal basis corresponding to the basis B = {(1, -1, 0), (1, 0, 3), (0, 1, 2)} for R^3.</td><td>CO3</td><td>(6)</td></tr>
<tr><td colspan="5">Module 4</td></tr>
<tr><td>15</td><td>a)</td><td>Determine whether the function $T : R^2 \rightarrow R^2$ defined by T(x, y) = (3x , x+1) is a linear transformation.</td><td>CO4</td><td>(3)</td></tr>
<tr><td></td><td>b)</td><td>Find a basis for the kernel and nullity of the linear transformation
$T : R^4 \rightarrow R^4$ defined by T(x) = Ax, where $x \in R^4$ and A=$\begin{bmatrix} 2 & -1 & 0 & 1 \\ 3 & 2 & 1 & -1 \\ 0 & 1 & -1 & 2 \\ 5 & 0 & -2 & 2 \end{bmatrix}$</td><td>CO4</td><td>(6)</td></tr>
<tr><td>16</td><td>a)</td><td>Let $T : R^2 \rightarrow R^3$ be a linear transformation such that T(1, 2) = (1, 0, 1), T(−1, 1) = (0, 1, -1). Find T(1, 1) and T(0, 0).</td><td>CO4</td><td>(4)</td></tr>
<tr><td></td><td>b)</td><td>Let $T : R^3 \rightarrow R^2$ be a linear transformation defined by
T (x, y, z) = (3x − 2z, 2y − z). Find the matrix for T relative to the bases B = {(1, 0, 1), (1, −1, 0), (0, 1, 1)} and C = {(1, 1), (1, 0)}.</td><td>CO4</td><td>(5)</td></tr>
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MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

SECOND SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code: GXEST203

Course Name: Foundations of Computing: From Hardware Essentials to Web Design

Max. Marks: 60		Duration: 2 hours 30 minutes

PART A

		Answer all questions. Each question carries 3 marks	CO	Marks
1		Explain the boot process of a computer.	CO1	(3)
2		Describe how the cache memory can enhance the performance of the CPU in a program execution environment.	CO1	(3)
3		Why is Unicode encoding better than ASCII? Is ASCII still important? Justify.	CO2	(3)
4		Explain how a CPU executes an arithmetic operation (e.g., addition of two numbers) using the fetch-execute cycle.	CO2	(3)
5		Consider a mutli-office company with each office site seperated by a few hundred kilometers. The offices are equipped with a network of computers. Explain the possible type of networking technolologies involved in the communication between any two computers of the company.	CO3	(3)
6		Is WWW same as the Internet? Justify.	CO3	(3)
7		Write a simple HTML code to display a heading and a paragraph.	CO4	(3)
8		In what situations would JavaScript be more useful than HTML and CSS alone? Provide an example.	CO4	(3)

PART B

Answer any one full question from each module. Each question carries 9 marks

Module 1

9	a)	Explain the memory hierarchy in a computer system with a labeled diagram and examples.	CO1	(5)
	b)	What are the advantages of using SSDs over HDDs for data storage?	CO1	(4)
10	a)	Draw a diagram to show the components of a motherboard. Explain the role of any three components in enabling effective system performance.	CO1	(6)
	b)	Explain the role of buses in data communication.	CO1	(3)
Module 2				
11	a)	Illustrate the addition of two 4 bit binary numbers. Explain the process step by step.	CO2	(5)
	b)	(i) Explain the function of the **Control Unit** within a CPU. (ii) Describe the function of an **instruction set** and explain how it facilitates software – hardware interaction.	CO2	(4)
12	a)	A new character encoding system is being developed for a multilingual application. Explain why Unicode would be preferred over ASCII for such purposes.	CO2	(5)
	b)	How does a computer represent the integer -12(negative 12) using 8-bit 2's complement notation? Show the steps involved in the conversion. Also, state the number of bits required to represent a value of 255 in binary.	CO2	(4)
Module 3				
13	a)	A company has multiple offices in different cities and requires seamless data sharing. Compare the suitability of client-server and peer-to-peer networks for this purpose, and recommend the better option.	CO3	(5)
	b)	Give the Linux commands for the following operations: (i) Create a directory (ii) List all the files and sub folders in a directory (iii) Copy the contents of a text file (iv) Change the access permissions of files and directories.	CO3	(4)
14	a)	A startup is expanding its business and needs to implement network security. Propose a security plan that includes desktop security, perimeter security, and VPN	CO3	(5)

		for remote access.		
	b)	Answer the following: (i) Explain the concept of a Client-Server network and provide a real-world example. (ii) Explain the role of DNS in internet communication.	CO3	(4)
Module 4				
15	a)	Design a webpage using HTML and CSS for an online bookstore. Include a title, a heading, a list of book categories, and a basic style.	CO4	(6)
	b)	Inspect the following XHTML and make the necessary corrections required, if any: <!DOCTYPE html> <html xmlns="https://ktu.edu.in/"> <head> </head> <BODY> <i>Some text</i> <P>This is a paragraph</P> <a HREF="https://www.ktu.edu.in/">Visit our university website </BODY>	CO4	(3)
16	a)	A university website needs to display a dynamic countdown timer for an upcoming event. Explain how JavaScript can be used to achieve this, with a sample script.	CO4	(6)
	b)	Illustrate the different ways in which style sheets can be applied to a web page with simple examples.	CO4	(3)
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MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
SECOND SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2024

Course Code: GXEST204
Course Name: PROGRAMMING IN C

Max. Marks: 60 | Duration: 2 hours 30 minutes

PARTA

		Answer all questions. Each question carries 3 marks	CO	Marks

1		Convert the below if-else to conditional operator
		`if (first_check)`
		`{`
		`"access denied";`
		`}`
		`else`
		`{`
		`if (second_check)`
		`{`
		`"access denied";`
		`}`
		`else`
		`{`
		`"access granted";`
		`}`
		`}`
			1	(3)

2		State proper reasons to identify whether the given identifiers are valid or not. i) 9marks ii) Mark 2020 iii) v4vote iv) _csdept v) @cse vi) AxB	1	(3)
3		Explain the different ways in which you can declare and initialize a single-dimensional array.	2	(3)
4		How can a list of strings be stored within a two-dimensional array? How can the individual strings be processed? What library functions are available to simplify string processing?	2	(3)
5		Suppose function F1 calls function F2 within a C program. Does the order of the function definitions make any difference? Explain.	3	(3)

6		Identify the storage class in the below code?
		`#include<stdio.h>`
		`int a;`
		`int main()`
		`{`
		`printf("a=%d",a);`
		`return 0;`
		`}`
			5	(3)

7		Suppose a function receives a pointer as an argument. Explain how the function	4	(3)

		prototype is written. In particular, explain how the data type of the pointer argument is represented.		
8		While using the statement, fp = fopen("myfile.c", "r") ; what happens if, − 'myfile.c' does not exist on the disk − 'myfile.c' exists on the disk	5	(3)
PARTB				
Answer any one full question from each module. Each question carries 9 marks				
Module 1				
9	a	Write a C Program to check if a given number is a strong number or not. A strong number is a number in which the sum of the factorial of the digits is equal to the number itself. Eg:- 145=1!+4!+5!=1+24+120=145	1	(5)
	b	Write a C program that calculates and prints the result of the following expression: result = 5 * (6 + 2) / 4 - 3. Explain how parentheses affect the precedence of operators in this expression.	1	(4)
10	a	Write a C program that reads employee's ID, name, status ('F' for full-time and 'P' for part-time) and basic pay. DA and HRA for full-time employees are 75% and 10% of basic pay respectively. Part-time employees have basic pay only. Print the total salary of the employee.	1	(6)
	b	Explain the difference between primitive (basic) data types and derived data types in C. Provide examples for each category. Why is it important to choose the correct data type for variables in a program?	1	(3)
Module 2				
11	a	Write a C program to reverse the content of an array without using another array.	2	(4)
	b	Given a list of cities, their literacy level, and the population, write a program to print the list cities sorted lexicographically. Assume duplication in city names.. Break the ties with the value of literacy level, and then the population.	2	(5)
12	a	Write a C program to rotate (re-insert rightmost element at the left) elements of an array to the right by n-steps.	2	(4)
	b	Write a C program that allows the user to enter a line of text, store it in an array, and then display it backward. The line length should be unspecified (terminated by pressing the Enter key), but assume it will not exceed 80 characters.	2	(5)
Module 3				
13	a	Develop a function that takes the two dates as input and compares them. The function should return 1 if date1 is earlier than date2, return 0 if date1 is later than date2. Write code to display whether date1 is earlier, later, or the same as date2 based on the function's result.	3	(4)
	b	Define a structure named Vehicle with members model (character array), year (integer), and price (float). Create an array of 10 Vehicle structures and write a function to accept user input for each vehicle's details and Sort the array in descending order based on price.	2	(5)
14	a	Define a structure that can describe a hotel with the following members: name, address, grade, average room charge, and number of rooms. Write a function to print out hotels of a given grade, sorted in order of their average room charges.	3	(4)
	b	Write a function to print out hotels with an average room charge less than a specified value in the above problem.	3	(5)

Module 4				
15	a	Write a c program snippet to open a file in write mode and check if the file opened successfully. if it fails ,print an error message and terminate the program.	5	(4)
	b	Two persons want to access a file "sample.txt". First person want to read the data from the file. The second person want to read and write the data from and to the file simultaneously. Can you help them to do so by writing the corresponding programming codes?	5	(5)
16	a	Write a function calculates the roots of a quadratic equation with coefficients *a*, *b*, and *c* (passed as pointer parameters).	4	(4)
	b	In a small firm, employee numbers are given in serial numerical order, that is 1, 2, 3, etc. − Create a file of employee data with following information: employee number, name, sex, gross salary. − If more employees join, append their data to the file. − If an employee with serial number 25 (say) leaves, delete the record by making gross salary 0. − If some employee’s gross salary increases, retrieve the record and update the salary. Write a program to implement the above operations.	5	(5)

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MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

SECOND SEMESTER B. TECH DEGREE EXAMINATION, APRIL 2025

Course Code: PCCST205

Course Name: Discrete Mathematics

Max. Marks: 60 Duration: 2 hours 30 minutes

Part A			
Answer all questions. Each question carries 3 marks.		**CO**	**Marks**
1	Each student in Liberal Arts at some college has a mathematics requirement M and a science requirement S. A poll of 140 students shows that 60 students have met the requirement M, 45 have met S, 20 have met both M and S. Determine how many students have met neither of the two requirements.	2	(3)
2	Let $\mathcal{P}(S)$ be the power set of some set S. If the join and meet operations are defined as union $(\cup)$ and intersection $(\cap)$ respectively, prove that $\{\mathcal{P}(S), \subseteq\}$ forms a lattice.	4	(3)
3	Write the inverse, converse, and contrapositive of the following conditional statement. *If I finish my homework before dinner and it does not rain, then I will go to the ball game.*	1	(3)
4	Prove that if n is a positive integer, then n is even if and only if $7n + 4$ is even.	1	(3)
5	Prove that $1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$.	1	(3)
6	A factory makes custom sports cars at an increasing rate. In the first month, only one car is made, in the second month two cars are made, and so on, with n cars made in the n^{th} month. Set up a recurrence relation for the number of cars produced in the first n months by this factory.	1	(3)
7	Show that the cube roots of unity form a cyclic group.	1	(3)
8	Let G be a group with subgroups H and K. If $\|G\| = 660, \|K\| = 66$, and $K \subset H \subset G$, what are the possible values for $\|H\|$?	1	(3)
Part B			
Answer any one full question from each module. Each question carries 9 marks.			
Module I			

9 (a)	Show that the set of all positive rational numbers is countably infinite.	1	(4)
(b)	Consider the relation $R = \{(1,1),(1,2),(2,1),(2,2),(3,3)\}$ on $S = \{1,2,3\}$. Show that R is an equivalence relation. Also find the partition defined by R on S.	1	(5)
10 (a)	Let $f, g, h : \mathbb{R} \to \mathbb{R}$, where $f(x) = x^2, g(x) = x+5, h(x) = \sqrt{x^2+2}$. Illustrate the associativity property of the composition of three functions using f, g and h.	1	(5)
(b)	Let A, B, C be three arbitrary sets. Show that $$(A - B) - C = A - (B \cup C).$$	1	(4)
Module II			
11 (a)	An island has two tribes of natives. Any native from the first tribe always tells the truth, while any native from the other tribe always lies. You arrive at the island and ask a native if there is gold on the island. He answers, "There is gold on the island if and only if I always tell the truth". Is there gold on the island?	1	(4)
(b)	Ten people volunteer for a three-person committee. Every possible committee of three that can be formed from these ten names is written on a slip of paper, one slip for each possible committee, and the slips are put in ten hats. Show that at least one hat contains 12 or more slips of paper.	1	(5)
12 (a)	Prove the validity of the following argument: *All hummingbirds are richly colored.* *No large birds live on honey.* *Birds that do not live on honey are dull in color.* ――――――――――――― $\therefore$ *Hummingbirds are small.*	1	(5)
(b)	Prove that if $m + n$ and $n + p$ are even integers, where m, n, and p are integers, then $m + p$ is even.	1	(4)
Module III			
13 (a)	Let $\langle a_n \rangle_{n \in \mathbb{N}}$ be a sequence satisfying $a_0 = 0$, and $$a_n = (n+1) \cdot a_{n-1} + n \quad \text{for all } n \geq 1.$$ Use induction to show that $a_n = (n+1)! - 1$ for all $n \in \mathbb{N}$.	2	(4)

(b)	Let $\langle a_n \rangle_{n\in\mathbb{N}}$ be a sequence satisfying $$a_n = 2a_{n-1} + 3a_{n-2} \text{ for all } n \geq 2$$ i. Given that a_0, a_1 are odd, prove that a_n is odd for all $n \in \mathbb{N}$. ii. Given that $a_0 = a_1 = 1$, prove that $a_n = \frac{1}{2}\left(3^n - (-1)^{n+1}\right)$ for all $n \in \mathbb{N}$.	5	(5)
14 (a)	When climbing a staircase, you can take either one, two, or three stairs in a single step. i. Write "Let S_n be the number of ways to climb a staircase with n stairs." Prove the recurrence relation $S_n = S_{n-1} + S_{n-2} + S_{n-3}$. ii. Write down what S_0, S_1, and S_2 are (note $S_0 = 1$ since there is one way to climb a staircase with zero stairs: do nothing). No justification needed. iii. Use (i) and (ii) to answer the following question. How many ways are there to climb a staircase with 9 stairs?	5	(4)
(b)	For integers $n \geq 2$, find and prove a formula for $$\prod_{i=2}^{n} \left(1 - \frac{(-1)^i}{i}\right).$$	2	(5)
Module IV			
15 (a)	Prove that every cyclic group is abelian.	6	(4)
(b)	Let G be a group. Let H be a subgroup of G and let N be a normal subgroup of G. The **product** of H and N is defined to be the subset $$H \cdot N = \{hn \in G \mid h \in H, n \in N\}.$$ Prove that the product $H \cdot N$ is a subgroup of G.	6	(5)
16 (a)	Let G, G' be groups. Let $\phi : G \to G'$ be a group homomorphism. Then prove that for any element $g \in G$, we have $$\phi(g^{-1}) = \phi(g)^{-1}.$$	6	(4)

(b)	Let G be a finite group. Let S be the set of elements g such that $g^5 = e$, where e is the identity element in the group G. Prove that the number of elements in S is odd.	6	(5)

MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

SECOND SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code : PCITT205

Course Name: DISCRETE MATHEMATICAL STRUCTURES

Max. Marks: 60 | Duration: 2 hours 30 minutes

PART A

		Answer all questions. Each question carries 3 marks	CO	Marks
1		Using truth table, verify the logical equivalence $[p \to (qvr)] \Leftrightarrow [\neg r \to (p \to q)]$	CO1	(3)
2		Define the terms: Converse, Inverse and Contrapositive.	CO1	(3)
3		What is the pigeonhole principle? Explain. If you select any five numbers from 1 to 8 then prove that at least two of them will add up to 9.	CO2	(3)
4		In how many ways can the letters of the word 'SOCIOLOGICAL' be arranged such that all the vowels are together?	CO2	(3)
5		Define poset with an example.	CO3	(3)
6		Let S = {0,1,2,3 4 …}. Is (S, +) a monoid?	CO4	(3)
7		Find the generating function for the sequence 1,1,1…,1.0,0,0…. where the first n+1 terms are 1	CO5	(3)
8		A bank pays 6% interest on savings, compounding the interest yearly. Write the recurrence relation for this and solve it to find how much will a deposit of Rs 1000 be worth after 12 years.	CO5	(3)

PART B

Answer any one full question from each module. Each question carries 9 marks

		Module 1		
9	a)	Without using truth tables, show that $(p \vee q) \wedge \neg(\neg p \wedge q) \Leftrightarrow p$	CO1	(5)
	b)	Let $p(x), q(x)$and $r(x)$ be the following open statements $p(x): x^2 - 7x + 10 = 0, q(x): x^2 - 2x - 3 = 0, r(x): x < 0$ Determine the truth or falsity of the following statements, where the universe is all integers. If a statement is false, provide a counter example or explanation.$(i) \forall x[p(x) \to \neg r(x)]$ $(ii) \forall x[q(x) \to r(x)]$ $(iii) \exists x[q(x) \to r(x)]$ $(iv) \exists x[p(x) \to r(x)]$)	CO1	(4)
10	a)	Establish the validity of the following argument	CO1	(5)

		$p \rightarrow (q \rightarrow r)$ $p \vee \neg s$ q ――――― $s \rightarrow r$		
	b)	Express the negation of the statement using quantifier "if the teacher is absent, then some students do not keep quiet."	CO1	(4)
		Module 2		
11	a)	Determine the number of integers n, $1 \leq n \leq 2000$ that are not divisible by 2,3 or 5.	CO2	(5)
	b)	Determine the number of integer solutions of $x_1 + x_2 + x_3 + x_4 = 32$, such that (i) $x_i \geq 0$, $1 \leq i \leq 4$ (ii) $x_i > 0$, $1 \leq i \leq 4$	CO2	(4)
12	a)	In how many ways can 10 identical dimes be distributed among five children if (a) there are no restrictions (b) each child gets at least one dime (c)the oldest child gets at least 2 dimes.	CO2	(5)
	b)	Determine the coefficient of $w^3x^2yz^2$ in $(2w - x + 3y - 2z)^8$	CO2	(4)
		Module 3		
13	a)	Show that set $Q^+$of all positive rational numbers forms an abelian group under the operation * defined by $a * b = \frac{ab}{2},\ a, b \epsilon Q^+$	CO4	(5)
	b)	Find the complement of each element in D_{42}.	CO3	(4)
14	a)	Consider the set A= {a, b, c}. Show that Q(A), the set of all proper subsets of A is a partially ordered set under the relation ⊆ , the set inclusion. Draw the Hasse diagram for the poset (Q(A), ⊆).	CO3	(5)
	b)	For $A = \{1,2,3\}$ and $B = \{w, x, y, z\}$. Give an example for a relation from A to B which is not a function.	CO3	(4)
		Module 4		
15	a)	Solve a_{n+1}-a_n=2n+3, $n \geq 0, a_0 = 1$.	CO5	(5)
	b)	Determine the sequence generated by the exponential generating function $f(x) = \frac{1}{1-x}$	CO5	(4)
16	a)	Solve the recurrence relation $2a_n = 7a_{n-1} - 3a_{n-2}, a_0 = 2, a_1 = 5$	CO5	(5)
	b)	Solve the recurrence relation $a_{n+1} = 2a_n + 1$, $n \geq 0$, $a_0 = 0$.	CO5	(4)

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MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

SECOND SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code: UCEST206

Course Name: ENGINEERING ENTREPRENEURSHIP AND IPR

Max. Marks: 40		Duration: 2 Hours 30 minutes

PART A

		Answer all questions. Each question carries 3 marks	CO	Marks
1		Define the term *entrepreneurial mindset* and explain its importance.		(2)
2		List and briefly explain the types of Intellectual Property Rights (IPR).		(2)
3		What is value proposition, and why is it important for startups?		(2)
4		Describe the role of customer feedback in refining market strategies.		(2)
5		Write the steps involved in preparing a financial projection for a business plan.		(2)
6		Briefly describe two stakeholder engagement strategies used in prototype development.		(2)

PART B

Answer any one full question from each module. Each question carries 7 marks

		Module 1		
7		Discuss the frameworks for fostering innovation in startups. How does building a strong team influence the success of innovation?		(7)
8		A startup aims to launch a mobile-based health tracking application. Identify the statutory compliances they need to fulfill and outline strategies for protecting their intellectual property.		(7)
		Module 2		
9		Explain the steps involved in conducting market segmentation and customer profiling. How does competitor analysis enhance a startup's value proposition?		(7)
10		A group of entrepreneurs is developing a solution canvas for a solar-powered water purifier targeting rural markets. Design their value proposition and draft a market validation strategy using customer feedback.		(7)
		Module 3		

11		What are the key components of a business plan? Highlight the role of marketing and operations strategies in ensuring business sustainability		(7)
12		A startup is launching a food delivery app focusing on local cuisine. Develop a business plan framework including market analysis, marketing strategy, and risk mitigation approaches		(7)
Module 4				
13		Describe the prototype development process, including resource allocation and quality assurance. Why is engaging with advisors and mentors important during this phase?		(7)
14		An engineering team is prototyping a low-cost robotic arm for industrial use. Create a stakeholder engagement strategy to attract investors and secure partnerships while ensuring product quality and timely delivery.		(7)
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Sample Questions- Module wise

Module-I: Introduction to Ideation, Innovation & Entrepreneurship

Total Marks: 30
Duration: 90 Mins

PART A: Short Answer Questions

(Answer all questions. Each question carries 3 marks.)
(6 x 3 = 18 Marks)

1. Define *innovation* and discuss its importance in entrepreneurship.
2. Differentiate between *copyrights* and *patents* in Intellectual Property Rights (IPR).
3. What are the statutory requirements for starting a business in India?
4. Explain the concept of *team dynamics* and its importance in entrepreneurship.
5. Identify two idea generation techniques and describe their application.
6. Discuss the role of IPR in securing funding for startups.

PART B: Descriptive and Case Study Questions

(Answer one question from each set. Each question carries 6 marks.)
(2 x 6 = 12 Marks)

7(Descriptive):
(a) Discuss the frameworks for innovation and their application in the engineering domain. *(4 marks)*
(b) Explain the significance of identifying pain points in developing innovative solutions. *(2 marks)*

OR

8 (Case Study):
A software engineer has developed a new AI tool to optimize inventory management for small businesses.
(a) Identify the types of IPR they can use to protect their innovation. *(2 marks)*
(b) Suggest a strategy to form a team and identify essential skill sets. *(2 marks)*
(c) Recommend how they can validate their idea with potential users. *(2 marks)*

9 (Descriptive):
(a) What is the entrepreneurial mindset, and how does it drive innovation? *(3 marks)*
(b) Explain strategies for refining ideas to bring innovation to life. *(3 marks)*

OR

10 (Case Study):
A group of students aims to start a drone delivery business.
(a) Identify the statutory requirements they need to comply with. *(2 marks)*
(b) Propose how thexy can validate their idea in the market. *(2 marks)*
(c) Discuss the importance of team roles in executing their plan effectively. *(2 marks)*

Module-II: Problem and Solution Canvas Preparation

Total Marks: 30
Duration: As per exam schedule

PART A: Short Answer Questions

(Answer all questions. Each question carries 3 marks.)
(6 x 3 = 18 Marks)

1. Define *value proposition* and explain its significance in entrepreneurship.
2. What is *customer segmentation*, and why is it important?
3. Outline the purpose of *persona development* in customer profiling.
4. Discuss the role of competitor analysis in market positioning.
5. Describe the process of conducting market research for a startup.
6. Explain the importance of customer feedback in refining a product.

PART B: Descriptive and Case Study Questions

(Answer one question from each set. Each question carries 6 marks.)
(2 x 6 = 12 Marks)

7 (Descriptive):
(a) Explain the steps involved in preparing a problem canvas. *(3 marks)*
(b) Discuss how regulatory and legal considerations influence market entry strategies. *(3 marks)*

OR

8 (Case Study):
A startup wants to create eco-friendly packaging solutions.
(a) Identify their potential competitors and suggest profiling methods. *(2 marks)*
(b) Conduct a SWOT analysis for their business idea. *(2 marks)*
(c) Propose a pricing strategy based on market research. *(2 marks)*

9 (Descriptive):
(a) What is customer needs assessment, and how does it help in product development? *(3 marks)*
(b) Explain the significance of competitive analysis in creating a differentiation strategy. *(3 marks)*

OR

10 (Case Study):
A food-tech company plans to launch a subscription-based meal delivery service.
(a) Describe how they can profile their target customers. *(2 marks)*
(b) Suggest ways to validate their market assumptions. *(2 marks)*
(c) Recommend strategies for effective communication and messaging. *(2 marks)*

Module-III: Business Plan Preparation

Total Marks: 30
Duration: As per exam schedule

PART A: Short Answer Questions

(Answer all questions. Each question carries 3 marks.)
(6 x 3 = 18 Marks)

1. Define *business plan* and its significance for startups.
2. List the components of a marketing and sales strategy in a business plan.
3. What is risk management, and why is it important?
4. Discuss the purpose of financial projections in a business plan.
5. Explain the role of iterative development in prototype creation.
6. What are the benefits of quality assurance in prototype development?

PART B: Descriptive and Case Study Questions

(Answer one question from each set. Each question carries 6 marks.)
(2 x 6 = 12 Marks)

7 (Descriptive):
(a) Discuss the key elements of a product/service description in a business plan. *(3 marks)*
(b) Explain the steps involved in preparing a prototype development plan. *(3 marks)*

OR

8 (Case Study):
A wearable tech startup wants to create a fitness tracking device.
(a) Develop an operations plan, including resource allocation. *(3 marks)*
(b) Identify key risks and propose mitigation strategies. *(3 marks)*

9 (Descriptive):
(a) Explain how market analysis informs a business plan. *(3 marks)*
(b) Discuss the importance of documentation and version control in prototype development. *(3 marks)*

OR

10 (Case Study):
A renewable energy company is planning to launch a wind energy solution for urban use.
(a) Suggest a marketing strategy for the product. *(2 marks)*
(b) Recommend a financial projection for their business plan. *(2 marks)*
(c) Describe a feedback loop for improving the product. *(2 marks)*

Module-IV: Prototype Development

Total Marks: 30
Duration: As per exam schedule

PART A: Short Answer Questions

(Answer all questions. Each question carries 3 marks.)
(6 x 3 = 18 Marks)

1. Define the phases of prototype development.
2. What are the key components of a stakeholder engagement strategy?
3. Explain the role of testing in prototype development.
4. How can iterative development improve a prototype?
5. Discuss the significance of mentor engagement in startup growth.
6. What is the importance of engaging investors during prototype development?

PART B: Descriptive and Case Study Questions

(Answer one question from each set. Each question carries 6 marks.)
(2 x 6 = 12 Marks)

7 (Descriptive):
(a) Describe the role of technical specifications in prototype development. *(3 marks)*
(b) Discuss methods for engaging stakeholders during product development. *(3 marks)*

OR

8 (Case Study):
A healthcare startup is designing a wearable device to monitor blood sugar levels.
(a) Propose a prototype development timeline. *(2 marks)*
(b) Suggest ways to test and ensure quality assurance for the prototype. *(2 marks)*
(c) Recommend methods to secure funding from investors. *(2 marks)*

9 (Descriptive):
(a) Explain the importance of resource allocation in prototype development. *(3 marks)*
(b) Discuss how feedback from partners and customers can refine a prototype. *(3 marks)*

OR

10 (Case Study):
An IoT company is creating a smart home automation system.
(a) Develop a strategy for engaging early adopters and collecting feedback. *(2 marks)*
(b) Propose a differentiation strategy for their product in the market. *(2 marks)*
(c) Suggest methods for maintaining version control during development. *(2 marks)*

MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

SECOND SEMESTER B. TECH DEGREE EXAMINATION, MAY 2025

Course Code: **UCEST206**

Course Name: Engineering Entrepreneurship and IPR

Max. Marks: 40	Duration: 2 hours 30 minutes

PART A

Read the following paragraph carefully and answer all questions following the text block. Each question carries 2 marks

A new startup called "EcoPack" has been created by a group of entrepreneurs with a focus on providing sustainable packaging solutions made from biodegradable materials. The team identified a growing concern in the market regarding single-use plastics and the environmental impact of traditional packaging. They developed a novel packaging material that decomposes quickly and is environmentally friendly. During their entrepreneurial journey, they were concerned about the possibility of their innovation being copied by others, so they made plans to protect both their technology and brand. The team then identified their target customers, primarily eco-conscious businesses and consumers, and understanding their pain points regarding plastic waste. They also developed targeted marketing strategies and customized their product for different size, thickness and having different strengths to better meet customer needs. To bring the idea to life, EcoPack developed a plan outlining their value proposition, market analysis, and financial projections. The final phase of product development involved rigorous testing and refinement of the biodegradable material, ensuring it met quality standards while being cost-effective. Throughout this process, EcoPack engaged investors, environmental organizations and potential customers, to gain feedback and refine their product offering.

PART A

No.	Question	CO	Marks
1	Discuss briefly the different ways EcoPack team could protect their idea of biodegradable packaging innovation?	CO1	2
2	Could you point out why the team behind EcoPack decided to make this product?	CO2, CO3	2
3	In what way did EcoPack customize its product offerings to better meet customer needs?	CO2, CO3	2
4	What key elements did EcoPack include in their business plan to outline their market positioning and financial projections?	CO4	2
5	How did EcoPack ensure that their biodegradable packaging met quality standards and was cost-effective during the product development phase?	CO5	2
6	What strategies did EcoPack use to engage stakeholders such as investors and customers during their product development process? Discuss in brief.	CO5	2

PART B			
Answer any one full question from each module. Each question carries 7 marks			
Module 1			
7	Discuss various frameworks for innovation and evaluate their practical applications in the development of a new product or service. Use relevant examples to illustrate the effectiveness of these frameworks in real-world scenarios.	CO1, CO2	7
8	Discuss strategies for protecting intellectual property, considering the different types of innovation. Provide practical scenarios where these strategies can be effectively applied to safeguard innovations in a competitive market.	CO1, CO2	7
Module 2			
9	Using the Problem and Solution Canvas, outline your approach to solving the real-world issue of reducing food waste in urban areas. Highlight the key steps involved in identifying the problem, defining objectives, and implementing strategies to address this challenge.	CO2, CO3	7
10	Perform a SWOT analysis for launching a startup in the electric vehicle (EV) market. Discuss how the strengths, weaknesses, opportunities, and threats identified in the analysis can inform market positioning and differentiation strategies for the startup.	CO2, CO3	7
Module 3			
11	Prepare a brief framework for a business plan for a new app-based service targeting fitness enthusiasts. Include key components such as market analysis, target audience, competitive landscape, value proposition, operations plan, revenue model, marketing strategy, and financial projections.	CO3, CO4	7
12	Outline a prototype development plan for a wearable health-monitoring device. Discuss the key phases, including requirements analysis, resource allocation, design specifications, and the iterative development process. Highlight the importance of testing and feedback loops in refining the prototype to meet user needs and technical standards.	CO3, CO4	7
Module 4			
13	Describe strategies for effective stakeholder engagement, focusing on how startups can attract and retain investors and advisors. Discuss approaches such as building trust, demonstrating value, and maintaining transparent communication. Provide examples of successful startup strategies in engaging stakeholders.	CO4, CO5	7
14	Explain the process of competitor profiling and benchmarking when launching a new product. Discuss how understanding competitors' strengths, weaknesses, and market positioning can guide pricing decisions and differentiation strategies to gain a competitive edge.	CO5	7